E. Knight Paths

time limit per test: 1 second
memory limit per test: 256 megabytes

A knight stands in the top-left corner of an 8×8 chessboard. You can move it **at most** k times. Count such possible paths modulo 2^{32}.

Formally, count paths of cells $(C_1, C_2, \ldots, C_s)$ such that $1 \le s \le k+1$ and a knight can move from C_i to C_{i+1} for every i.

(A chess knight moves to a square that is two squares away horizontally and one square vertically, or one away horizontally and two vertically.)

Input
An integer k $(0 \le k \le 10^9)$.

Output
Print the answer modulo $2^{32} = 4294967296$.

Examples

input
```
1
```
output
```
3
```

input
```
2
```
output
```
15
```

input
```
6
```
output
```
17231
```

Note
In the first sample test, a knight can either stay in the initial cell $(1,1)$ or move to $(2,3)$ or $(3,2)$. That's 3 ways in total.

Matrix Exponentiation

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
336283821	Aug/30/2025 19:19	Happy New Year!
336183761	Aug/29/2025 22:38	Happy New Year!
336183690	Aug/29/2025 22:37	Delivered to wrong address on flight 14 WA14